

4.4 Taylor Theorem Worksheet

1. Use an appropriate Taylor polynomial of degree 2 to obtain a good estimate for $\sqrt{403}$.

2. Consider the function $f(x) = e^{-x}$.

a) Use an appropriate Taylor polynomial of degree 2 to obtain a good estimate for $e^{-1/2}$.

b) Use Taylor's Theorem to find a bound on the error of this approximation.

c) Use Taylor's Theorem to show that $f(x) = e^{-x}$ is equal to its Taylor's series, centered at x_0 , for *all* x .

d) How many terms of the Taylor series centered at 0 are needed to estimate $e^{-1/2}$ accurately to 3 decimal places?

3. The function $f(x) = \cosh(x)$ has a Taylor series about 0 that is $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$.
- a) Use the first 3 terms of this series to obtain an approximation for $\cosh(1)$.

b) Use Taylor's Inequality to determine an upper bound for the error of this approximation.

c) How many terms of the Taylor series are needed to estimate $\cosh(1)$ accurately to 5 decimal places?